

Dr. Satyvir Singh

Scientific Staff

Institute for Applied and Computational Mathematics, RWTH Aachen University
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Nationality: Indian | **Work Authorization:** German Permanent Resident (Niederlassungserlaubnis)

RESEARCH INTERESTS

Applied mathematics, Scientific computing, Computational fluid mechanics, High-order numerical methods, Hydrodynamic instabilities, Gas kinetic theory, Artificial intelligence, and Machine learning.

EXPERIENCE

- Scientific Staff at **Institute for Applied and Computational Mathematics, RWTH Aachen University, Germany** (Sept 2022–Present)
 - Course instructor: Mathematical Models for PDEs at **Institute for Applied and Computational Mathematics, RWTH Aachen University, Germany** (Apr 2023–Jul 2023)
 - Research fellow at **School of Physical and Mathematical Sciences, Nanyang Technological University, Singapore** (Aug 2018–July 2022)
 - Course tutor at **School of Physical and Mathematical Sciences, Nanyang Technological University, Singapore** (Aug 2021–Nov 2021)
 - Senior researcher at **Research Center for Aircraft Core Technology, Gyeongsang National University, South Korea** (Mar 2018–Jul 2018)
 - Visiting research student at **School of Power and Energy, Northwestern Polytechnical University, Xi'an, China** (July 2015–Aug 2015).
 - Research student at **School of Mech. & Aerospace Engineering, Gyeongsang National University, South Korea** (Mar 2013–Feb 2018).
 - Assistant Professor in **Department of Mathematics, Galgotias College of Engineering & Technology, Greater Noida, India** (Oct 2011–Feb 2013).
 - Guest lecturer in **Vivekananda College, Delhi University, New Delhi, India** (Jul 2012–Dec 2012).
 - Assistant Professor in **Department of Computer Sciences & Engineering, IMS Engineering College, Ghaziabad, India** (Jul 2011–Oct 2011).
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EDUCATIONAL QUALIFICATIONS

Mar 2013–Feb 2018: Ph.D. in Engineering

Aerospace Computational Modeling Laboratory,
School of Mech. & Aero. Engineering, Gyeongsang National University, South Korea
Thesis title: Development of a 3D discontinuous Galerkin method for the second-order Boltzmann-Curtiss-based hydrodynamic models of diatomic and polyatomic gases.

Aug 2009–Jun 2011: M.Tech. in Industrial Mathematics & Scientific Computing

Department of Mathematics,
Indian Institute of Technology Madras, Chennai, India.
Thesis title: A complete heatline analysis on mixed convection within a square cavity: Effect of aspect ratio and thermal boundary conditions.

July 2003–Jun 2005: Master of Science (M.Sc.)
Department of Mathematics,
Chaudhary Charan Singh University, Meerut, India.

July 2000–Jun 2003: Bachelor of Science (B.Sc.)
Chaudhary Charan Singh University, Meerut, India.
Subjects: Mathematics, Chemistry, and Physics

AWARDS/FELLOWSHIPS AND OTHER EXAMINATIONS

- Brain Korea 21+ Scholarship for Ph.D. Program at Gyeongsang National University, South Korea (2013–2016).
- GATE Financial MHRD Scholarship for M.Tech. Program at Indian Institute of Technology Madras, Chennai, India (2009–2011).
- Qualified CSIR-JRF National Eligibility Test (NET) in Mathematical Sciences (June 2011) with All India Rank #38.
- Qualified Graduate Aptitude Test in Engineering (GATE)-2012 with All India Rank #240

SKILLS

- Programming & Software Engineering: Python, C++, Fortran, Julia, Git, CI/CD (GitLab)
- CFD Tools & Meshing: OpenFOAM, Trixi.jl, ANSYS Fluent, Athena++, Gmsh, Pointwise
- Numerical Techniques: Discontinuous Galerkin, Finite Volume, Finite Element
- Application Areas: Compressible Flows, Shock Wave Dynamics, Multiphase Flows, Rarefied Gas Dynamics
- Heat Transfer & Multiphase Systems: Natural & mixed convection, Newtonian & Non-Newtonian fluid flows.
- Post-processing & Documentation: Tecplot, ParaView, LaTeX, Gnuplot, MS Word

KEY CFD PROJECTS

- **Postdoctoral Research Project**, RWTH Aachen University, Germany
 - Developed and implemented advanced computational models (Grad-type) for non-equilibrium gas flows.
 - Conducted high-fidelity simulations to analyze gas flow behavior around complex aerodynamic geometries.
 - Collaborated with international research teams to enhance CFD methodologies for aerospace applications.
- **Postdoctoral Research Project**, Nanyang Technological University, Singapore
 - Developed 2D DG solver to solve Boltzmann Transport Equation under electronic nonequilibrium conditions.
 - Developed efficient computational algorithms to simulate shock-induced flow instabilities.
 - Published findings in high-impact journals and presented at international conferences.
- **Postdoctoral Research Project**, Research Center for Aircraft Core Technology, South Korea
 - Enhanced computational efficiency and robustness of high-fidelity rarefied flow simulations.

- Implemented Langmuir and Maxwell-Smoluchowski boundary conditions for accurate rarefied gas flow simulations.
 - **Ph.D. Research Project**, Gyeongsang National University, South Korea
 - Developed a 3D mixed modal DG solver for simulating rarefied and microscale gas flows.
 - Implemented scalable computational techniques for solving complex fluid dynamics problems.
 - **M.Tech. Research Project**, Indian Institute of Technology Madras, India
 - FEM-based mixed convection analysis → applied to cooling system design methodologies.
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PUBLICATIONS

Preprints Submitted

- Song, H., **Singh, S.**, Cayci, S., Torrilhon, M. (2026), Extraction of moment closures for strongly non-equilibrium flows via machine learning (under review in *Physics of Fluids*).

Published Journals

- Ankur, Jiware, R., and **Singh, S.** (2026), Finite element method for the numerical simulation of modified Poisson-Nernst-Planck/Navier-Stokes model, *Applied Numerical Mathematics*, **223**:255–278.
- **Singh, S.**, Karchani, A. (2026), Amplitude impact on single-mode Richtmyer–Meshkov flow in a stratified heavy fluid layer, *Physics of Fluids*, **38**:022105.
- Habscheid, J. **Singh, S.**, Theisen, L., Braun, S. and Torrilhon, M. (2026), A finite element solver for a thermodynamically consistent electrolyte model, *Computer Physics Communications*, **319**:109916.
- Msmali, A.H., **Singh, S.**, Ahmadini, A.A.H. (2026), Numerical Study of Amplitude-Driven Flow Dynamics in Shocked Heavy-Fluid Layers, *Mathematics*, **14**:82.
- **Singh, S.**, and Torrilhon, M. (2025), Shock-accelerated dynamics of heavy rectangular bubbles: Influence of aspect ratio on Richtmyer–Meshkov instability, *Physics of Fluids*, **37**:122129.
- Alsaeed, S.S., and **Singh, S.** (2025), Shock-driven flow physics of Richtmyer–Meshkov instability in tandem light cylindrical interfaces, *Physics of Fluids*, **37**:122116.
- Alsaeed, S.S., **Singh, S.**, Alshammari, N.F. (2025), Shock Mach number effect on instability evolution at a light-heavy fluid layer: A numerical investigation, *Axioms*, **14**:813.
- Alsaeed, S.S., **Singh, S.** (2025), A mixed local discontinuous Galerkin scheme for capturing shock-driven instabilities in compressible multicomponent flows, *AIMS Mathematics*, **10**:26389–26417.
- Alsaeed, S.S., **Singh, S.**, and Alrubea, N.A. (2025), Layer thickness impact on shock-accelerated interfacial instabilities in single-mode stratifications, *Applied Sciences*, **15**:10687.
- Alsaeed, S.S., **Singh, S.**, and Alrubea, N.A. (2025), Numerical Investigation of Atwood number effects on shock-driven single-mode stratified heavy fluid layers, *Mathematics*, **13**:3032.
- **Singh, S.**, Msmali, A. H., Tamsir, M., and Ahmadini, A.A.H (2025), Mechanisms of coupling-induced instabilities in shock-accelerated tandem light square bubbles, *Physics of Fluids*, **37**:082101.
- Msmali, Alsaeed, A. H., **Singh, S.**, and Meetei, M. Z. (2025), Single-Mode Richtmyer–Meshkov Instability in Light Fluid Layer: Insights from Numerical Simulations, *Axioms*, **14**:473.
- Song, H., **Singh, S.**, and Torrilhon, M. (2025), Non-equilibrium flow simulations based on Grad-14 and Grad-17 moment equations for polyatomic gases, *Physics of Fluids*, **37**:036140.
- Zhang, S. B., **Singh, S.**, Torrilhon, M., Zhang, H. H., Chen, Z.H., and Zheng, C. (2025), Numerical investigation of Richtmyer–Meshkov instability in shock-driven light square bubble via magnetohydrodynamics, *Computers and Fluids*, **299**:106698.

- Alsaeed, S. S, and **Singh, S.** (2025), Insights into coupling effects of double light square bubbles on shocked hydrodynamic instability, *Physica D: Nonlinear Phenomena*, **476**: 134646.
- Ankush, Sengupta, B., **Singh, S.**, and Lawrence Raj, P. R. (2025), Role of bulk viscosity on the flow physics past a rotating cylinder, *Physics of Fluids*, **37**:015147.
- Alsaeed, S. S., and **Singh, S.** (2024), Numerical study of thermal non-equilibrium effects on Richtmyer-Meshkov flow driven by a heavy forward-triangular bubble, *Physical Review E*, **100**:065103.
- Alsaeed, S. S., and **Singh, S.** (2024), Computational study on flow characteristics of shocked light backward-triangular bubbles in polyatomic gases, *Axioms*, **13**:843.
- Sreekala, V. A., Sengupta, B., Kammara, K. K., and **Singh, S.** (2024), On the evolution of magnetohydrodynamic flow instability in shock-accelerated light bubble, *Physical of Fluids*, **36**:106140.
- **Singh, S.**, and Alsaeed, S. S. (2024), High-fidelity simulations of Richtmyer-Meshkov flows triggered by a forward-pentagonal bubble with different Atwood numbers, *European Journal of Mechanics/B Fluids*, **108**:151–165.
- Alsaeed, S. S., and **Singh, S.** (2024), Computational Study of Shocked V-Shaped N₂/SF₆ Interface across Varying Mach Numbers, *Axioms*, **13**:700.
- Bansal, S., Kumar, A., Saini, A., Negi, A. S., and **Singh, S.** (2024), Exploring the Atwood number impact on shock-driven hydrodynamic instability at pentagonal interface using discontinuous Galerkin simulations, *Physica D: Nonlinear Phenomena*, **467**:134276.
- Alsaeed, S. S., and **Singh, S.** (2024), Modal Discontinuous Galerkin Simulations of Richtmyer–Meshkov Instability at Backward-Triangular Bubbles: Insights and Analysis, *Mathematics*, **12**:2005.
- **Singh, S.**, Msmali, A. H., and Nelson, M. I. (2024), Unfolding of shocked hydrodynamic instability at SF₆ elliptical interface: Physical insights from numerical simulations, *Computers & Fluids*, **277**:106304.
- Alsaeed, S. S., and **Singh, S.** (2024), Numerical Study of Shock Wave Interaction with V-Shaped Heavy/Light Interface, *Mathematics*, **12**:3131.
- **Singh, S.**, and Msmali, A. H. (2024), Analyzing Richtmyer-Meshkov phenomena triggered by forward-triangular light gas bubbles: A numerical perspective, *Mathematics*, **13**(6):365.
- **Singh, S.**, Song, H., and Torrilhon, M. (2024), Modal discontinuous Galerkin simulations for Grad’s 13-moment equations: Application to Riemann problem in continuum–rarefied flow regime, *Journal of Computational and Theoretical Transport*, 398–422.
- **Singh, S.**, Sengupta, B., and Rana, S. (2024), Analysis of Thermal Mixing and Entropy Generation during Natural Convection Flows in Arbitrary Eccentric Annulus, *Axioms*, **13**:233.
- **Singh, S.**, Sengupta, B., Awasthi, M. K., and Kumar, V. (2024), Insight on the flow physics of shock-driven elliptical gas inhomogeneity with different Atwood numbers, *International Journal of Mathematical, Engineering and Management Sciences*, **9**(1):1–22.
- **Singh, S.**, and Jalleli, D. T. (2024), Investigation of coupling effect on the evolution of Richtmyer–Meshkov instability at double heavy square bubbles, *SCIENCE CHINA Physics, Mechanics & Astronomy*, **67**(1):214711.
- **Singh, S.**, and Msmali, A. H. (2023), On the spatiotemporal pattern formation in nonlinear coupled reaction-diffusion systems, *Axioms*, **12**(11):1004.
- **Singh, S.** (2023), Investigation of aspect ratio effects on flow characteristics and vorticity generation in shock-induced rectangular bubble, *European Journal of Mechanics/B Fluids*, **101**:131–148.
- **Singh, S.**, and Battiato, M. (2023), Investigation of shock Mach number effects on the growth of convergent Richtmyer Meshkov instability in a heavy square bubble, *Physica D: Nonlinear Phenomena*, **453**:133844.

- **Singh, S.**, Mittal, R. C., Thottoli, S. R., and Tamsir, M. (2023), High-fidelity simulations for Turing pattern formation in multi-dimensional Gray-Scott reaction-diffusion system, *Applied Mathematics and Computation*, **452**:128079.
- **Singh, S.**, and Torrilhon, M. (2023), On the shock-driven hydrodynamic instability in square and rectangular light gas bubbles: a comparative study from numerical simulations, *Physics of Fluids*, **35**:012117.
- Shukla, A. K., Awasthi, M. K., and **Singh, S.** (2023), Impact of heat and mass transport on Rayleigh-Taylor instability of Walter's B viscoelastic fluid layer, *Microgravity Science and Technology*, **35**(3):1–11.
- **Singh, S.**, and Battiato, M. (2022), Numerical simulations of Richtmyer-Meshkov instability of SF6 square bubble in diatomic and polyatomic gases, *Computers & Fluids*, **242**:105502.
- **Singh, S.**, Karchani, A., Chourushi, T., and Myong, R. S. (2022), A three-dimensional modal discontinuous Galerkin method for second-order Boltzmann-Curtiss constitutive models of rarefied and microscale gas flows, *Journal of Computational Physics*, **457**:111052.
- Chourushi, T., Rahimi, A., **Singh, S.**, Ejethadi, O., Mankodi, T. K., and Myong, R. S. (2022), Thermal and flow characteristics of nonequilibrium monatomic, diatomic, and polyatomic gases in a cylindrical Couette flow based on the second-order non-Navier-Fourier constitutive model, *International Journal of Heat and Mass Transfer*, **187**:122580.
- Chourushi, T., **Singh, S.**, Vishnu, A. S., and Myong, R. S. (2022), Computational study of hypersonic rarefied gas flow over re-entry vehicles using the second-order Boltzmann-Curtiss constitutive model, *International Journal of Computational Fluid Dynamics*, **35**(8):566–593.
- **Singh, S.** (2021), Contribution of Mach number to the evolution of Richtmyer-Meshkov instability induced by a shock-accelerated square light bubble, *Physical Review Fluids*, **6**:104001.
- **Singh, S.** (2021), Mixed-type discontinuous Galerkin approach for solving the generalized FitzHugh-Nagumo reaction-diffusion model, *International Journal of Applied and Computational Mathematics*, **7**:207.
- **Singh, S.**, (2021), Numerical investigation of thermal non-equilibrium effects of diatomic and polyatomic gases on the shock-accelerated light square bubble using a mixed-type modal discontinuous Galerkin method, *International Journal of Heat and Mass Transfer*, **179**:121708.
- **Singh, S.**, Battiato, M., and Myong, R. S. (2021), Impact of bulk viscosity on flow morphology of shock-accelerated cylindrical light bubble in diatomic and polyatomic gases, *Physics of Fluids*, **33**:066103.
- **Singh, S.**, and Battiato (2021), An explicit modal discontinuous Galerkin method for Boltzmann transport equation under electronic nonequilibrium conditions, *Computers & Fluids*, **224**:104972.
- **Singh, S.**, and Battiato, M. (2021), Behavior of a shock-accelerated heavy cylindrical bubble under non-equilibrium conditions of diatomic and polyatomic gases, *Physical Review Fluids*, **6**:044001.
- **Singh, S.** (2020), Role of Atwood number on flow morphology of a planar shock-accelerated square bubble: A numerical study, *Physics of Fluids*, **32**:126112.
- **Singh, S.**, and Battiato, M. (2020), Strongly out-of-equilibrium simulations for electron Boltzmann transport equation using explicit modal discontinuous Galerkin method, *International Journal of Applied and Computational Mathematics*, **6**:133.
- **Singh, S.**, and Battiato, M. (2020), Effect of strong electric fields on material responses: The Bloch oscillation resonance in high field conductivities, *Materials*, **13**(5):1070.
- **Singh, S.**, Karchani, A., Sharma, K., and Myong, R. S. (2020), Topological representation of the second-order constitutive model based on Boltzmann-Curtiss kinetic equation for diatomic and polyatomic gases, *Physics of Fluids*, **32**:026104.

- Chourushi, T., Rahimi, A., **Singh, S.**, and Myong, R. S. (2020), Computational simulation of near continuum external gas flows using the Navier-Stokes-Fourier equations with slip/jump conditions based on a modal discontinuous Galerkin method, *Advances in Aerodynamics*, **2(8)**:1–37.
- **Singh, S.**, Karchani, A., and Myong, R. S. (2018), Nonequilibrium effects of diatomic and polyatomic gases on the shock-vortex interaction based on the second-order constitutive model of Boltzmann-Curtiss kinetic equation, *Physics of Fluids*, **30**:016109.
- Chourushi, T., **Singh, S.**, and Myong, R. S. (2018), Computational Study of Rarefied flow inside a lid-driven cavity using a mixed modal discontinuous Galerkin Method, *Journal of Computational Fluids Engineering*, **23(3)**:62–71.
- Raj, L. P., **Singh, S.**, Karchani, A., and Myong, R. S. (2017), A super-parallel mixed Galerkin method for the second-order Boltzmann based constitutive models of rarefied and microscale gases, *Computers & Fluids*, **57**:146–163.
- **Singh, S.**, and Myong, R. S. (2017), A computational study of bulk viscosity effects on shock vortex interaction using discontinuous Galerkin method, *Journal of Computational Fluids Engineering*, **22(2)**:86–95.

Conference Proceedings

- **Singh, S.**, and Torrilhon, M. (2026), Application of Regularized 13-Moment Equations to Continuum–Rarefied Gas Flow Over Aerospoke Blunt Body, *Springer Aerospace Technology*, **347–355**, 33th International Symposium on Rarefied Gas Dynamics, Göttingen, Germany, 2024.
- Hong, S., **Singh, S.**, and Torrilhon, M. (2026), Numerical Study of Grad-14 and 17 Moment Equations for Rarefied Polyatomic Gases, *Springer Aerospace Technology*, **379–387**, 33th International Symposium on Rarefied Gas Dynamics, Göttingen, Germany, 2024.
- Chourushi, T., **Singh, S.**, Vishnu, A. S., and Myong, R. S. (2024), Numerical simulations of rarefied gas flow over an aero-spiked hypersonic blunt body using the second-order Boltzmann-Curtiss constitutive model, *AIP Conference Proceedings*, **2996**, 140001, 32nd International Symposium on Rarefied Gas Dynamics, Seoul, South Korea, July 4–8, 2022.
- **Singh, S.**, and Battiato, M. (2022), Explicit modal discontinuous Galerkin approximations for three-dimensional electronic Boltzmann transport equation, *Lecture Notes in Computational Science and Engineering* **137**, Spectral and High Order Methods for Partial Differential Equations ICOSAHOM 2020+1, Vienna, Austria, July 12–16, 2021.
- **Singh, S.** (2022), A mixed-type modal discontinuous Galerkin approach for solving nonlinear reaction-diffusion equations, *AIP Conference Proceedings*, **2481**, 040037, International Conference on Advancements in Engineering and Sciences, Dehradun, India, July 1–2, 2021.
- **Singh, S.**, and Myong, R. S. (2019), Three-dimensional discontinuous Galerkin method for the second-order Boltzmann-Curtiss constitutive model for continuum-rarefied gas flows, *AIP Conference Proceedings*, **2132**, 060013, 31st International Symposium on Rarefied Gas Dynamics, University of Strathclyde, Glasgow, UK, July 23–27, 2018.
- Xiao, H., Myong, R. S., and **Singh, S.** (2014), A new near-equilibrium breakdown parameter based on the Rayleigh-Onsager dissipation function, *AIP Conference Proceedings*, **1628**, 527, 29th International Symposium on Rarefied Gas Dynamics, Xian, China, July 13–18, 2014.

Edited Books

- Awasthi, M. K., Kumar, A., Dutt, N., and **Singh, S.** (2024), Computational Fluid Flow and Heat Transfer: Advances, Design, Control, and Applications, *CRC Press*; ISBN: 9781032603186.
- **Singh, S.**, and Awasthi, M. K. (2026), Complex Engineering Systems - Modeling and Optimization, *De Gruyter Publisher*; ISBN: 9783111723464.
- **Singh, S.**, and Awasthi, M. K. (2026), Mathematical Solutions for Complex Engineering Systems, *River Publications*; ISBN: 9788743806691.

Book Chapters

- **Singh, S.** (2025) Dynamics of shock-accelerated V-shaped gas interface, in [Flow Dynamics and Heat Transfer: Computational Fluid Dynamics, Thermal Analysis, Numerical Simulations](#), De Gruyter (ISBN: 9783111660592).
- **Singh, S.** and Sengupta, B. (2024) Shock wave effects on hydrodynamic instability in elliptical bubbles, in [Computational Fluid Flow and Heat Transfer: Advances, Design, Control, and Applications](#), CRC Press (ISBN: 9781032603186).
- Rana, S., **Singh, S.** and Sengupta, B. (2024) Natural convection study in cylindrical annulus through OpenFOAM, in [Computational Fluid Flow and Heat Transfer: Advances, Design, Control, and Applications](#), CRC Press (ISBN: 9781032603186).
- **Singh, S.** (2023) Numerical Investigation of Wave Pattern Evolution in Gray-Scott Model Using Discontinuous Galerkin Finite Element Method, in [Advances in Mathematical and Computational Modeling of Engineering Systems](#), CRC Press (ISBN: 9781003367420).
- **Singh, S.**, and Awasthi, M. K. (2023) Mach number impact on Richtmyer-Meshkov instability in shock-refrigerant-22 bubble interaction, in [Advances in Mathematical and Computational Modeling of Engineering Systems](#), CRC Press (ISBN: 9781003367420).
- **Singh, S.** (2022) An Explicit Modal Discontinuous Galerkin Approach for Compressible Multicomponent Flows: Application to Shock-Bubble Interaction, in [Computing and Simulation for Engineers](#), CRC Press (ISBN: 9781003222255).
- **Singh, S.** (2022) Computational Modeling of Nonlinear Reaction-Diffusion Fisher-KPP Equation with Mixed Modal Discontinuous Galerkin Scheme, in [Mathematical Modeling for Intelligent Systems: Theory, Methods, and Simulation](#), CRC Press (ISBN: 9781003291916).

Conference Proceedings (Abstract Only)

- 5th European Conference on Non-Equilibrium Gas Flows (NEGF26), Toulouse, France (2026).
- 9th International Conference on Advanced Computational Methods in Engineering and Applied Mathematics (ACOMEN2025), Ghent, Belgium (2025).
- SIAM Conference on Computational Science and Engineering (CSE25), Fort Worth, Texas, USA (2025).
- 5th International Conference on Computational Engineering (ICCE 2024), Darmstadt, Germany (2024).
- High-Order Nonlinear Numerical Methods for Evolutionary PDEs (HONOM 2024), Chania, Crete Island, Greece (2024).
- 33rd International Symposium on Rarefied Gas Dynamics (RGD33), Göttingen, Germany (2024).
- 1st European Fluid Dynamics Conference (EFDC1), Aachen, Germany (2024).
- International Conference on Applied Mathematics and Mechanics (ICAMM 2023), IIT Indore, India (2023).
- 2023 SIAM Conference on Computational Science and Engineering (CSE23), Amsterdam, Netherlands (2023).
- 32nd International Symposium on Rarefied Gas Dynamics (Virtual Conference), South Korea (2022).
- International Conference on Spectral High Order Methods (Virtual Conference), Vienna, Austria (2021).
- International Conference on Advancements in Engineering and Sciences (ICAES2021), Graphic Era Hill University, Dehradun, India (2021).

- Pre-RGD32 Online Workshop, South Korea (2021).
- North American High Order Methods Conference (NAHOMCon), San Diego State University, USA (2019).
- IPS Annual Meeting 2019, Singapore University of Technology and Design, Singapore (2019).
- 31th International Symposium on Rarefied Gas Dynamics (RGD31), Glasgow, UK (2018).
- 1st National Symposium on Shock Waves (NSSW) Korea, Seoul National University, South Korea (2018).
- 2017 Asia-Pacific International Symposium on Aerospace Technology, (APISAT-2017) Seoul Olympic Parklet, South Korea (2017).
- 12th International Symposium on Mechanics, Aerospace and Informatics Engineering (ISMAI-2017) Goeje, South Korea (2017).
- 12th World Congress on Computational Mechanics (WCCM XII), Seoul, South Korea (2016).
- 9th National Congress of Fluids Engineering (9th NCFE), Seoul, South Korea (2016).
- 10th International Symposium of Mechanics, Aerospace and Informatics Engineering (ISMAI), Jinju, South Korea (2015).

INVITED TALKS

- Institut für Theoretische Physik, Ruhr-Universität Bochum, Germany (2023).
- Department of Mathematics, VIT-AP, India (2023).
- Department of Mathematics, Pandit Deendayal Energy University, Gandhinagar, India (2022).
- International Conference on Innovation and Application in Science and Technology (ICIAST-2021), Greater Noida, India (2021).
- Condensed Matter Physics Seminar, Nanyang Technological University, Singapore (2019).

STUDENTS' SUPERVISION

- Undergraduates
 1. Ms. Liwen Shen, Computational Engineering Science (CES), RWTH Aachen University
- Ph.D.
 1. Mr. Hang Song (ongoing: joined on Feb 2023), RWTH Aachen University
 2. Mrs. Seetu Rana (ongoing: joined on Aug 2022), GJUS&T Hisar, India

SPONSORED PROJECTS

- International consultant for the project entitled “Mathematical Modelling and High-fidelity Simulations for Brain Tumor Dynamics” funded by the Deanship of Graduate Studies and Scientific Research Scheme, Jazan University, Saudi Arabia (Grant: RG24-M05; 40,000 SAR) Sanctioned Year: 2024.

PROFESSIONAL ACTIVITIES

Guest Editor

- Physics of Fluids (AIP, 2026)
- Axioms (MDPI, 2025)
- Journal of Computational and Theoretical Transport (Taylor & Francis, 2024)

Editorial Board Member

- Scientific Reports (Springer)

Reviewer

- Physics of Fluids, Computers & Fluids, Shock Waves, International Journal of Heat and Fluid Flow, Journal of Applied Physics, Mathematical Methods in the Applied Sciences, Numerical Methods for Partial Differential Equations, International Journal of Applied and Computational Mathematics, Engineering Computations, Aerospace Systems, Applied Sciences

MEMBERSHIPS

- Society for Industrial and Applied Mathematics (SIAM): 2022 and 2025

REFERENCES

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